Machine Maintenance Insight — Start-to-Finish Assessment Guide

U23AD491 — Data Science | DS Sprint — Skill Based Assessment 2 | Use Case 25

Business question: What operating patterns indicate machines that require maintenance attention?

Critical data note: The provided question paper describes five CSV schemas but does not include actual operational records. This package contains synthetic data for the workflow. Never present synthetic results as verified MechaCare factory findings. If official CSVs are supplied, replace the raw files and rerun the notebook.

Deliverables checklist

• Cleaned/prepared data: cleaned/sensor_machine_analysis.csv, cleaned/machine_weekly_analysis.csv, cleaned/machine_maintenance_summary.csv

• Python notebook: MechaCare_EDA_Feature_Engineering.ipynb

• Excel workbook: MechaCare_Maintenance_Analysis.xlsx

• Business pitch: MechaCare_Business_Insight_Pitch.pptx

• Raw input tables: raw/sensor.csv, raw/machines.csv, raw/usage.csv, raw/alarms.csv, raw/maintenance.csv

Round 1 — Quiz (20 marks)

Revise the purpose of EDA; data types; missing values and duplicates; mean, median, standard deviation, quartiles and IQR; normalization versus standardization; categorical encoding; correlation versus causation; train/test split and data leakage; Pandas read_csv/info/isna/duplicated/groupby/merge; NumPy arrays and vectorization; Matplotlib/Seaborn histograms, boxplots, bar charts, line charts and scatterplots; Excel AVERAGE, MEDIAN, COUNT, COUNTIF, SUMIF, IF, IFERROR, XLOOKUP/VLOOKUP and Pivot Tables.

Round 2 — EDA and feature engineering (30 marks)

1. Load and inspect. Read the five CSVs. Record row/column counts, dtypes, missing values, exact duplicates and duplicate keys. Parse all timestamps with pandas.to_datetime(errors='coerce').

2. Clean carefully. Deduplicate sensor rows using machine + timestamp; standardize alarm labels; retain original sensor values and add outlier flags. Do not delete extreme values without checking whether they are genuine.

3. Join at the correct grain. Join sensor readings to machines (many readings to one machine). Aggregate sensor and alarm events to machine-week before joining weekly usage. Analyze maintenance at service-event or machine level. Do not merge all raw tables on machine alone.

4. Analyze six business questions. Abnormal sensor ranges; readings during the 21 days before service; most frequent alarms; load versus downtime; service burden by machine type; repeated service/warning patterns.

5. Engineer features. Weekly sensor means/maxima, number of sensor readings, outlier counts, alarm count, high-load flag, threshold-based sensor flags, warning score, service-event count, repeat-service flag and known downtime.

6. Explain limitations. Thresholds are provisional; missing downtime is unknown; missing alarm matches do not prove zero alarms; a repeated service event is not automatically a confirmed failure; correlation is not causation.

Round 3 — Excel (30 marks)

• Open the provided workbook and begin with Summary. Inspect sensor_clean, weekly_analysis, maintenance_summary, alarm_frequency, type_comparison, before_service and priority_ranking.

• Use filters, freeze panes, readable headers and number formats. Keep each record at a clearly defined grain.

• Use descriptive formulas: AVERAGE, MEDIAN, MIN, MAX, STDEV.S, COUNTIF, SUMIF/AVERAGEIF and IF. For missing values, avoid treating blank downtime as zero at the event level.

• Create or refresh Pivot Tables for alarm type frequency, service action frequency, machine type versus service count, and machine versus repeated service. Create Pivot Charts and a single summary sheet.

• Minimum visual evidence: sensor distributions; before-service trend; alarm frequency; load versus downtime; machine-type comparison; repeated-failure/service patterns.

Round 4 — Business pitch (20 marks)

Use 5–6 slides: (1) business question/objective, (2) data quality and method, (3) evidence-backed findings, (4) charts, (5) prioritized recommendations, (6) limitations and decision requested. For every finding, state the metric, what it means operationally, and what action management should take. Replace every synthetic numeric claim with results from official data if it becomes available.

Recommended maintenance decision logic

Prioritize a machine for engineering review when several independent signals persist together: abnormal vibration/temperature/current, recurring alarms, repeated service events and high known downtime. Use the priority score in this package as an explainable screening heuristic only—not as a validated failure prediction model. Confirm limits with maintenance engineers before issuing automated alerts.

Suggested 45-second pitch

“Our objective was to identify operating patterns that may indicate maintenance attention. We checked data quality, standardized alarm labels, flagged unusual sensor readings, and analyzed sensor and usage information at machine-week level to avoid inflating counts. We compared warning patterns with service history and known downtime, then ranked machines for review using transparent indicators. The recommendation is to inspect machines with persistent multi-sensor warnings and repeat service first, improve completeness of downtime and failure-cause records, and validate alert thresholds with engineers. These results must be confirmed against official operational data before production decisions.”

Package use

Extract the ZIP. Open the notebook from the package root and run it top to bottom. The notebook expects the raw folder to be beside it. Review generated tables and charts, then update the workbook and pitch. If the assessment platform provides official CSVs, replace the matching files in raw/ and rerun; do not keep the synthetic findings.